

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
SECOND SEMESTER 2021 – 2022
DEPARTMENT OF COMPUTER SCIENCE

Questions: 6

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Midsem Exam CLOSED BOOK

WEIGHTAGE : 30% (30 Marks)

DATE : 23-March-2022

DURATION : 90 Minutes (1:00 pm- 2:30 pm)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q. 1

Consider a Table 1 which shows average per capita wine consumption and heart disease rate in a random sample of 8 countries.

Table 1: Average per capita wine consumption vs heart disease rate

Yearly wine consumption in liters (x)	2.5	3.9	2.9	2.4	2.9	0.8	9.1	2.7
Death from heart diseases (y)	221	167	131	191	220	297	71	172

- i. Estimate the equation of regression line $y = mx + c$, for given data by considering yearly wine consumption in litres as independent variable and death from heart diseases as dependant variable. Show all steps of calculation. (3 M)
- ii. Draw the regression line that best fit the given data. Show all steps of calculation. (1 M)
- iii. Calculate the R^2 value for given data. Show all steps of calculation. (1 M)

5 M

Q. 2

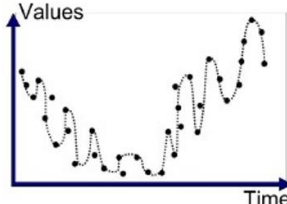
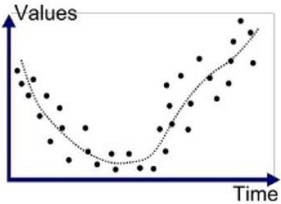
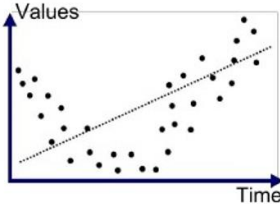
The Criscent lab is working on developing classification model for classifying viral and bacterial infection using CRP and temperature data. The data consider for the analysis is as follows,

Table 2: CRP and Temperature data for viral and bacterial infection

Infection	CRP	Temperature
Viral	40	36.0
Viral	11.1	37.2
Viral	30	36.5
Viral	21.4	39.4
Viral	10.7	39.6
Viral	3.4	40.7
Bacterial	42	37.6
Bacterial	31.1	42.2
Bacterial	50.0	38.5
Bacterial	60.4	39.4
Bacterial	45.7	38.6
Bacterial	17.3	42.7

5 M

	i. The Crisent Lab decided to use Fisher Linear Discriminant Analysis for the initial data processing. Calculate the between class scatter matrix, to expediate the process of classification model development. Show all steps of calculation. (3 M) ii. Calculate the discriminant scores if values of eigen vectors are, $V1 = 0.11$, $V2 = 0.70$. Also plot the scores in reduced dimension. Show all steps of calculation. (2 M)																																																			
Q. 3	The Crisent Lab is still struggling for developing a model for classifying viral and bacterial infection using data given in Table 2 of Q.2, they are not satisfied with performance by applying logistic regression on discriminant score and further they decided to use other method for classification on original data (without using discriminant scores). i. Which other classification technique will be suitable for the available data to better classify the infection into viral and bacterial? [Consider data given in Table 2 of Q.2] (1 M) ii. What will be the class label for test data, if CRP and temperature values of test data are 25 and 36 respectively? Show all steps of decision of class label. [Consider data given in Table 2 of Q.2 for training phase] (4 M)	5 M																																																		
Q. 4	Consider the following data related to different planets whether they are habitable for humans or not. Table 3: Data of planet habitability <table><tr><th>Planet Number</th><th>Size</th><th>Orbit</th><th>Temperature</th><th>Habitable</th></tr><tr><td>1</td><td>Large</td><td>Far</td><td>205</td><td>No</td></tr><tr><td>2</td><td>Large</td><td>Near</td><td>205</td><td>No</td></tr><tr><td>3</td><td>Large</td><td>Near</td><td>260</td><td>Yes</td></tr><tr><td>4</td><td>Large</td><td>Near</td><td>380</td><td>Yes</td></tr><tr><td>5</td><td>Small</td><td>Far</td><td>205</td><td>No</td></tr><tr><td>6</td><td>Small</td><td>Far</td><td>260</td><td>Yes</td></tr><tr><td>7</td><td>Small</td><td>Near</td><td>260</td><td>Yes</td></tr><tr><td>8</td><td>Small</td><td>Near</td><td>380</td><td>No</td></tr><tr><td>9</td><td>Small</td><td>Near</td><td>380</td><td>No</td></tr></table> i. What is the entropy of Habitable? Show all steps of entropy calculation. (1 M) ii. Which attribute will be chosen as the root of decision tree by considering the data shown in Table 3. (Note: Habitable as dependent variable and size, orbit, and temperature as independent variables)? Show all steps of decision of root node and also draw first level of decision tree. (4 M)	Planet Number	Size	Orbit	Temperature	Habitable	1	Large	Far	205	No	2	Large	Near	205	No	3	Large	Near	260	Yes	4	Large	Near	380	Yes	5	Small	Far	205	No	6	Small	Far	260	Yes	7	Small	Near	260	Yes	8	Small	Near	380	No	9	Small	Near	380	No	5 M
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8	Small	Near	380	No																																																
9	Small	Near	380	No																																																

Q.5	Consider the following confusion matrices obtained after classification of three types of fruits, apple, orange, and mango. <table><tr><td></td><td>Apple</td><td>Orange</td><td>Mango</td></tr><tr><td>Apple</td><td>35</td><td>9</td><td>11</td></tr><tr><td>Orange</td><td>13</td><td>43</td><td>8</td></tr><tr><td>Mango</td><td>9</td><td>10</td><td>24</td></tr></table> Calculate the respective (individual) F1-score and accuracy for class Apple, Orange and Mango. Show all steps of calculation.		Apple	Orange	Mango	Apple	35	9	11	Orange	13	43	8	Mango	9	10	24	5 M
	Apple	Orange	Mango															
Apple	35	9	11															
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Mango	9	10	24															
Q.6	i. What is difference between semi-supervised learning and active learning? ii. How regularization prevents overfitting? iii. How to find values of θ which will help to reduce cost across iterations? iv. Which of the following is overfitting, underfitting, and optimal model? Also, which of the following model will have high bias and high variance? (2 M) 	5 M																

***** ALL THE BEST *****

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SECOND SEMESTER 2021 – 2022
DEPARTMENT OF COMPUTER SCIENCE

Questions: 10

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Compre Exam CLOSED BOOK

WEIGHTAGE : 40% (40 Marks)

DATE : 8-June-2022

DURATION : 3 hrs (12:30 pm- 3:30 pm)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q.1

You are given a design matrix $X = \begin{bmatrix} 2 & 1 \\ 3 & 5 \\ 4 & 3 \\ 5 & 6 \\ 6 & 7 \\ 7 & 8 \end{bmatrix}$. Use Principal Component Analysis (PCA) to reduce the dimension of matrices from 2 to 1. Show all steps of calculation of covariance matrix, eigen values, normalized eigen vectors, and sample points in one dimension with respect to Principal Component 1.

5 M

Q.2

Write minimum three difference between following,
 a) K-means clustering vs EM for Gaussian mixture clustering.
 b) Supervised vs. unsupervised learning.

3 M

Q.3

You have six training points (A, B, C, D, E, F) and five classifiers (H1, H2, H3, H4, H5, H6) which make the following misclassifications:

Classifiers	Misclassified training points
H1	A, D, F
H2	D
H3	B, C
H4	A, B, F
H5	B, C, D
H6	A, C

Perform two rounds of boosting with these classifiers and training data. In each round, pick the classifier with the lowest error rate. Break ties by picking the classifier that comes first in this list: H1, H2, H3, H4, H5, H6. Show all steps of calculations and fill the following table,

	Round 1	Round 2
Weight _A		
Weight _B		
Weight _C		
Weight _D		
Weight _E		
Weight _F		

		Error rate of H1			
		Error rate of H2			
		Error rate of H3			
		Error rate of H4			
		Error rate of H5			
		Error rate of H6			
		Weak classifier (h)			
		Classifier error (ϵ)			
		voting power (α)			

Q. 4 Consider a following artificial neural network (as shown in Figure 2) with three inputs (x_1, x_2, x_3), two hidden neurons (H_1, H_2), two output neurons (O_1, O_2). The neuron network produces the output y_1 and y_2 . **3 M**

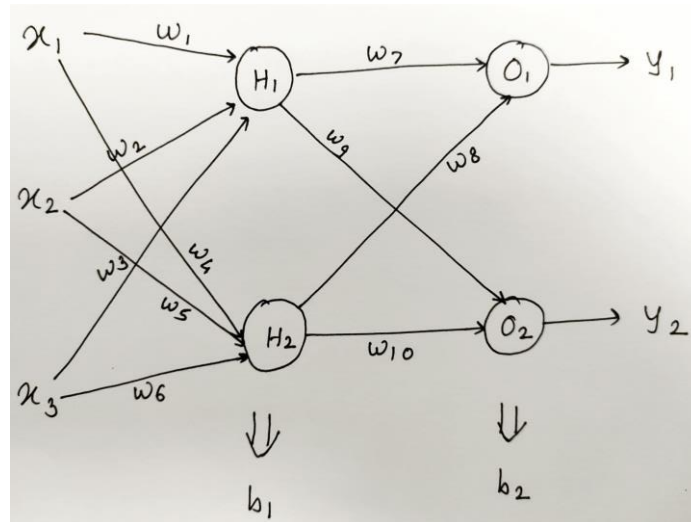


Figure 1 Q3: Artificial neural network

Here, w_1, w_2, w_3, w_4, w_5 , and w_6 are weights to hidden layer and w_7, w_8, w_9 , and w_{10} are weights to output layer. b_1 is bias to hidden layer and b_2 is bias to output layer.

Write the expressions for pre-activation ($\text{Pre}_{H1}, \text{Pre}_{H2}$) and activation (A_{H1}, A_{H2}) at hidden neurons, pre-activation ($\text{Pre}_{O1}, \text{Pre}_{O2}$) and activation (A_{O1}, A_{O2}) at output neurons. Consider the sigmoid as activation function for every neuron.

Q. 5 Consider the following plotted sample data points, as shown in Figure 3. **5 M**

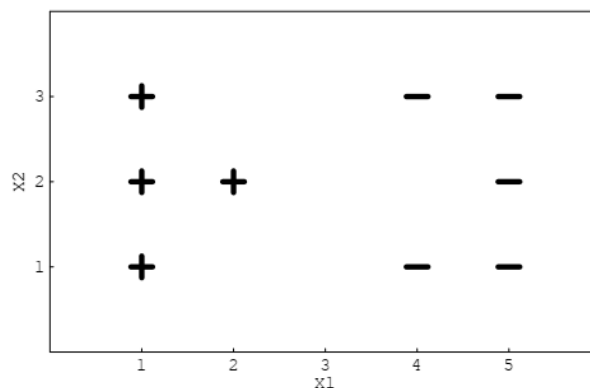


Figure 2 Q5: Sample data points

Find the support vectors and draw the decision boundary for linear SVM. (Show all steps of calculating discriminating hyperplane)

Q.6

Answer the following questions

3 M

- a) How many two class classifiers will be required to classify the sample points into 5 classes, using One-versus-one method?
- b) Consider the following plotted sample points as shown in Figure 5 and are using K-nearest neighbors method for classifying the sample points into *positive(+)* and *negative(-)*. Why might using too large values K be bad in this dataset? Why might too small values of K also be bad?

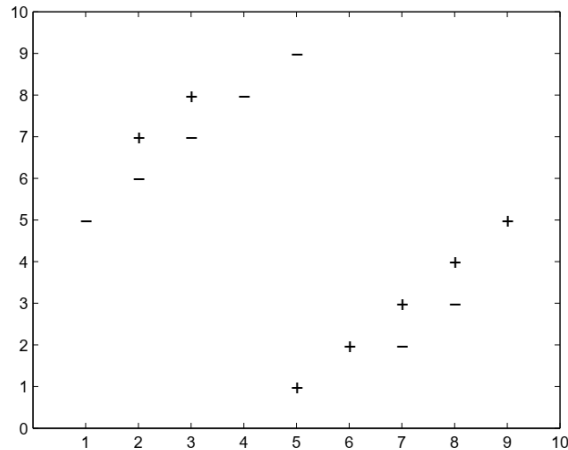


Figure 3 Q10 Sample data points

Q.7

Consider the Markov model with three states, 1, 2, and 3, that has the following transition probability matrix,

3 M

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

- a) Draw the state transition diagram for Markov model.
- b) If we know $P(X_1 = 1) = P(X_2 = 2) = 0.25$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

Q.8

Consider that a patient can have a symptom (S) that can be caused by two different diseases (A and B). It is known that the variation of gene G plays a big role in the manifestation of disease A. The Bayes' Net and corresponding conditional probability tables for this situation are shown below in Figure 4.

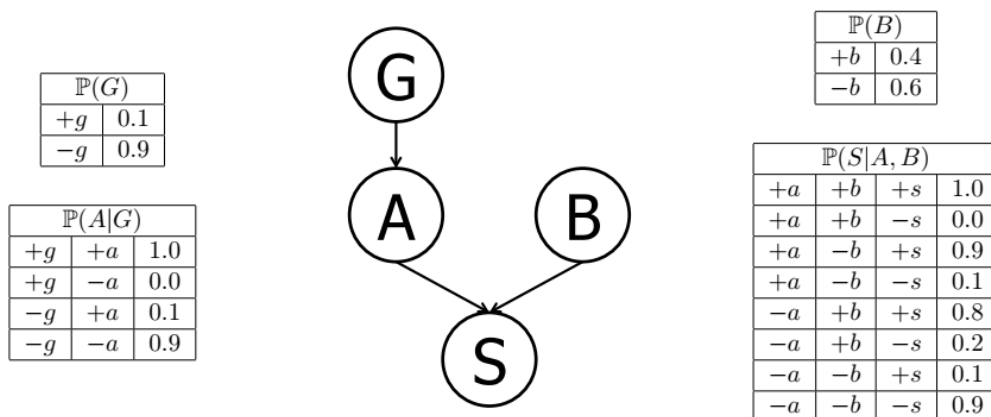
5 M

Figure 4 Q7: Bayes' net and corresponding probability tables

- a) What is the probability that a patient has disease A? (1 M)

- b) What is the probability that a patient has disease A given that they have symptom S and disease B? (2 M)
- c) What is the probability that a patient has the disease carrying gene variation G given that they have disease A? (2 M)

Q. 9 Consider a following Markov Decision Process (MDP) with rewards (r) for each state (as shown in Figure 1) and discount factor $\gamma = 0.5$, **3 M**

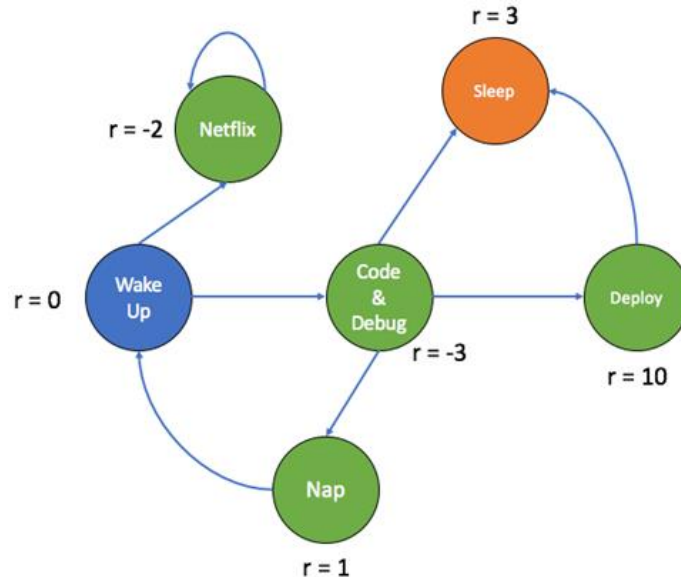


Figure 5 Q1: Markov decision process

- a) Calculate the accumulated rewards for path *WakeUp -> Code & Debug -> Nap -> WakeUp -> Netflix*. (Show all steps of calculations)
- b) Find the optimum policy (policy with maximum accumulated reward) for moving from WakeUP state to Sleep state. (Show all steps of calculations)

Q.10 Consider the following dataset of doses taken for vaccine and recovered from disease. **5 M**

No of doses (X)	Recovered from disease (Y)
4	0
5	0
3	1
2	1
2	0
1	1

The data scientist decided to use logistic regression model to take more insights into data and trying to find answers of following questions. (Show all steps of calculations)

- a) Calculate the probability of recovering from disease if person have taken 4 doses of vaccine.
- b) At least how many doses the person should take that make sure that he will recover from disease with probability of 90%?

***** ALL THE BEST *****